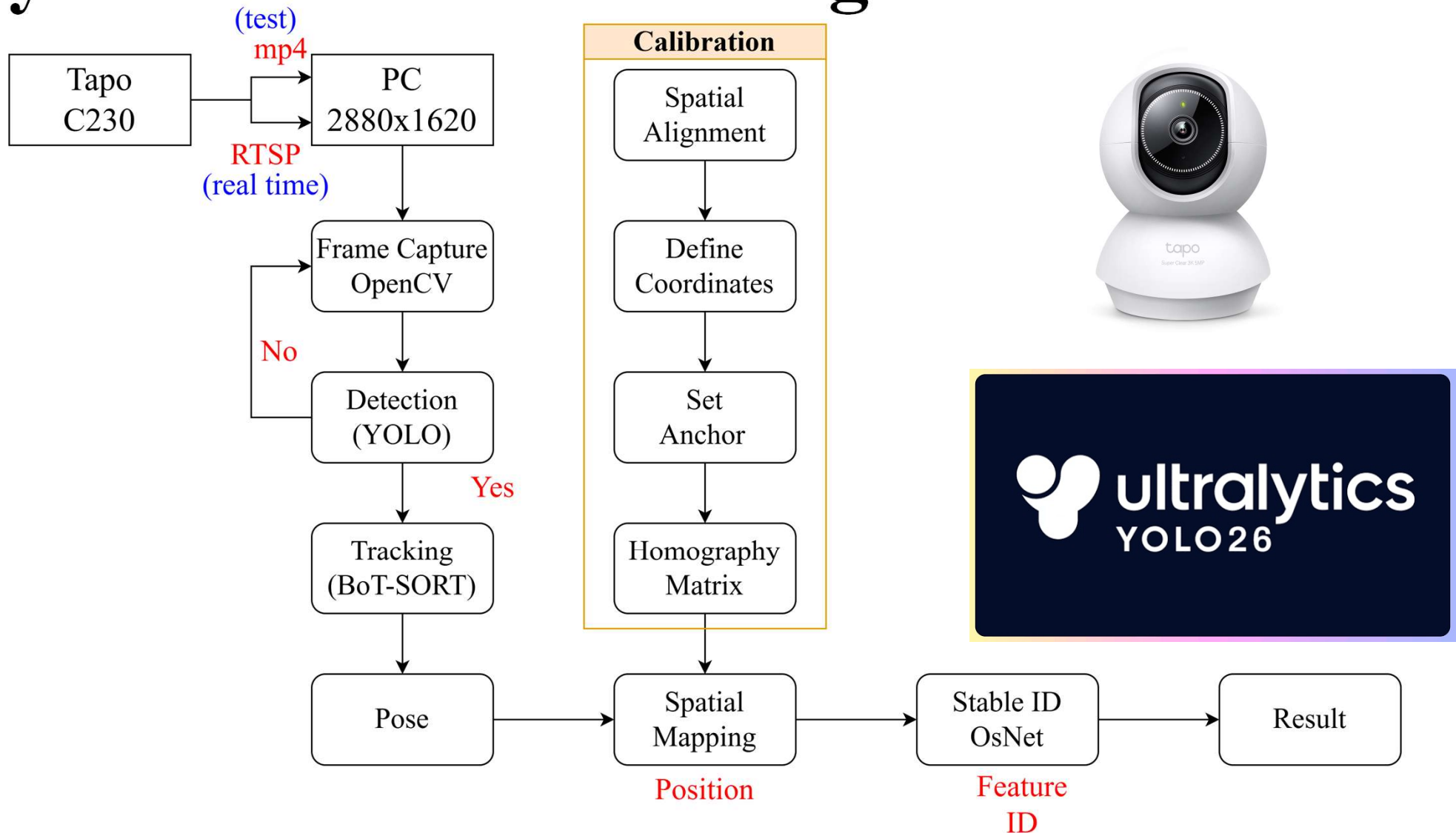


Person Detection & Tracking

日期:2026/9/18 報告人：陳眾科

System Architecture Diagram



Input Data

.Mp4 file

--stride = 5

Video : 20 fps

Ex: 1, 6 ,11, 16

原本是1秒20張畫面，跳幀後1秒用約4張畫面。

RTSP (real time)

只處理最新幀 (LatestFrameCapture)

Video : 20 fps

Ex: 0ms 、 50ms 、 100ms 、 150ms 、 200ms

1

2

3

4

5

start(1) 等待 等待 finish(1) start(5)

類似frame skip，不過沒有固定時間，根據電腦下運算。

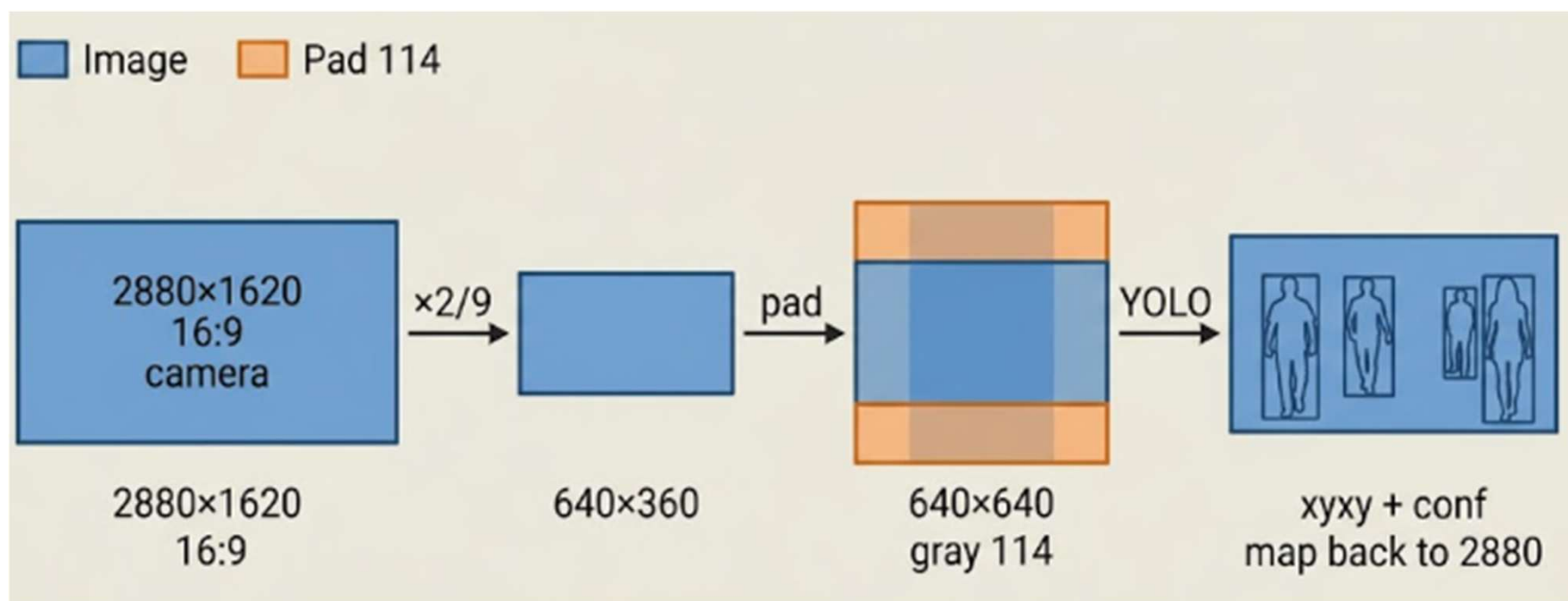
Detection(YOLO 26s)

只辨識COCO資料裡的Class 0 (人)

原始圖片大小: 2880 * 1620 (16:9)

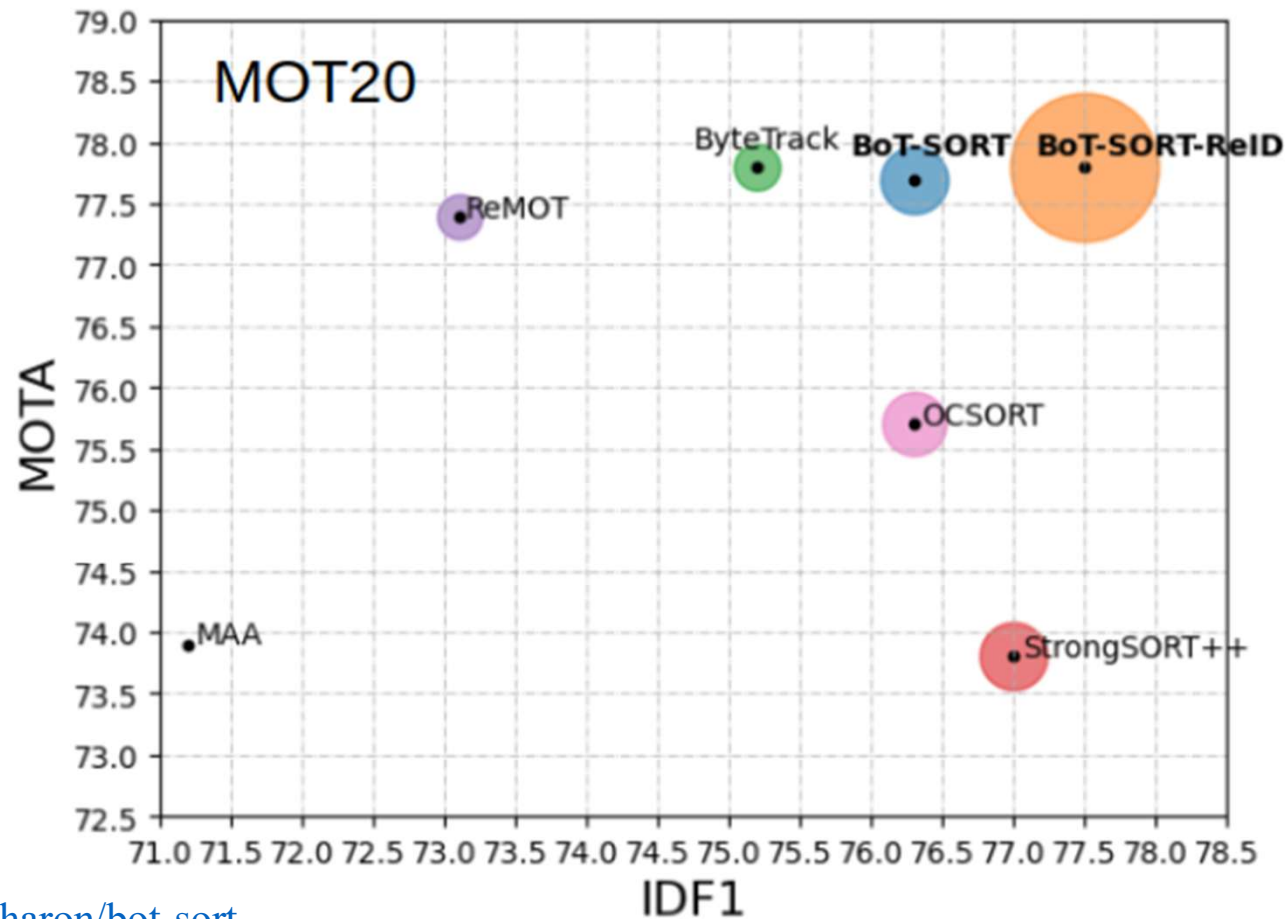
Resize後: 640 * 360 , 並補灰色在上下, 最終為640 * 640 。

得到框的座標 + 信心分數。



BoT-SORT

Best for: general-purpose tracking, especially moving cameras. Add ReID only when look-alike crowds cause ID swaps.



<https://github.com/niraharon/bot-sort>

YOLO26-pose

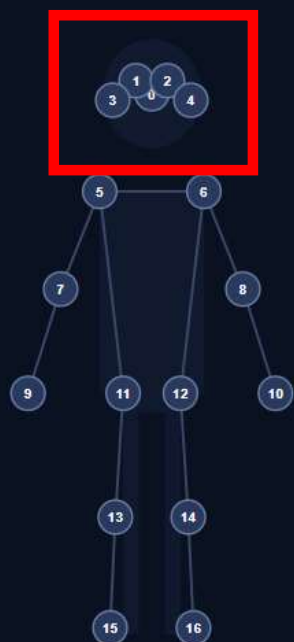
YOLO26-pose is pretrained on the **COCO Keypoints dataset**, which defines **17 keypoints** for human pose estimation. These keypoints cover the major joints and facial landmarks:

COCO-17 Keypoint Mapping

DATASET
COCO Keypoints

TOTAL POINTS
17

CATEGORIES
Face, Arms, Torso, Legs



0 Nose

2 Right Eye

4 Right Ear

6 Right Shoulder

8 Right Elbow

10 Right Wrist

12 Right Hip

14 Right Knee

16 Right Ankle

1 Left Eye

3 Left Ear

5 Left Shoulder

7 Left Elbow

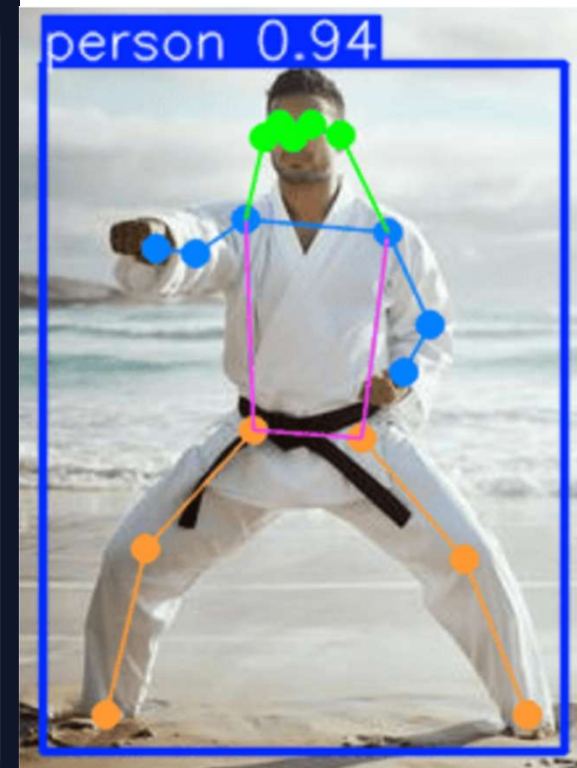
9 Left Wrist

11 Left Hip

13 Left Knee

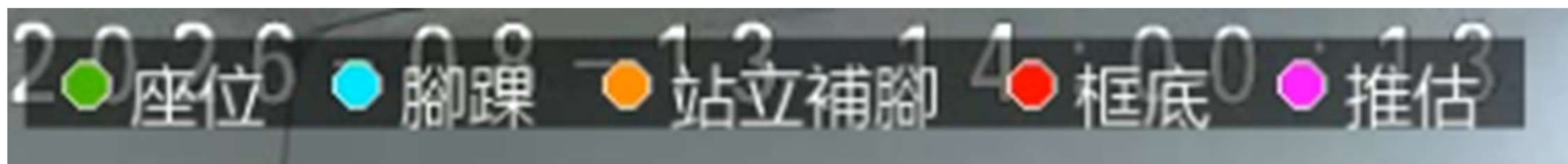
15 Left Ankle

17 keypoints covering facial landmarks and major body joints — the standard mapping used by YOLO26-pose and COCO-format models.



<https://learnopencv.com/yolo26-pose-estimation-tutorial/>

YOLO26-pose



標籤	意思
座位	左右髖的中點
腳踝	pose 直接用到腳踝
站立補腳	站著但腳被擋，用身體比例補
框底	直接用偵測框底邊
推估	頭頂／上半身下推估腳點（裁切時）

Calibration-(Spatial Alignment)

<https://github.com/ChenChungKo/person-detection-and-tracking>

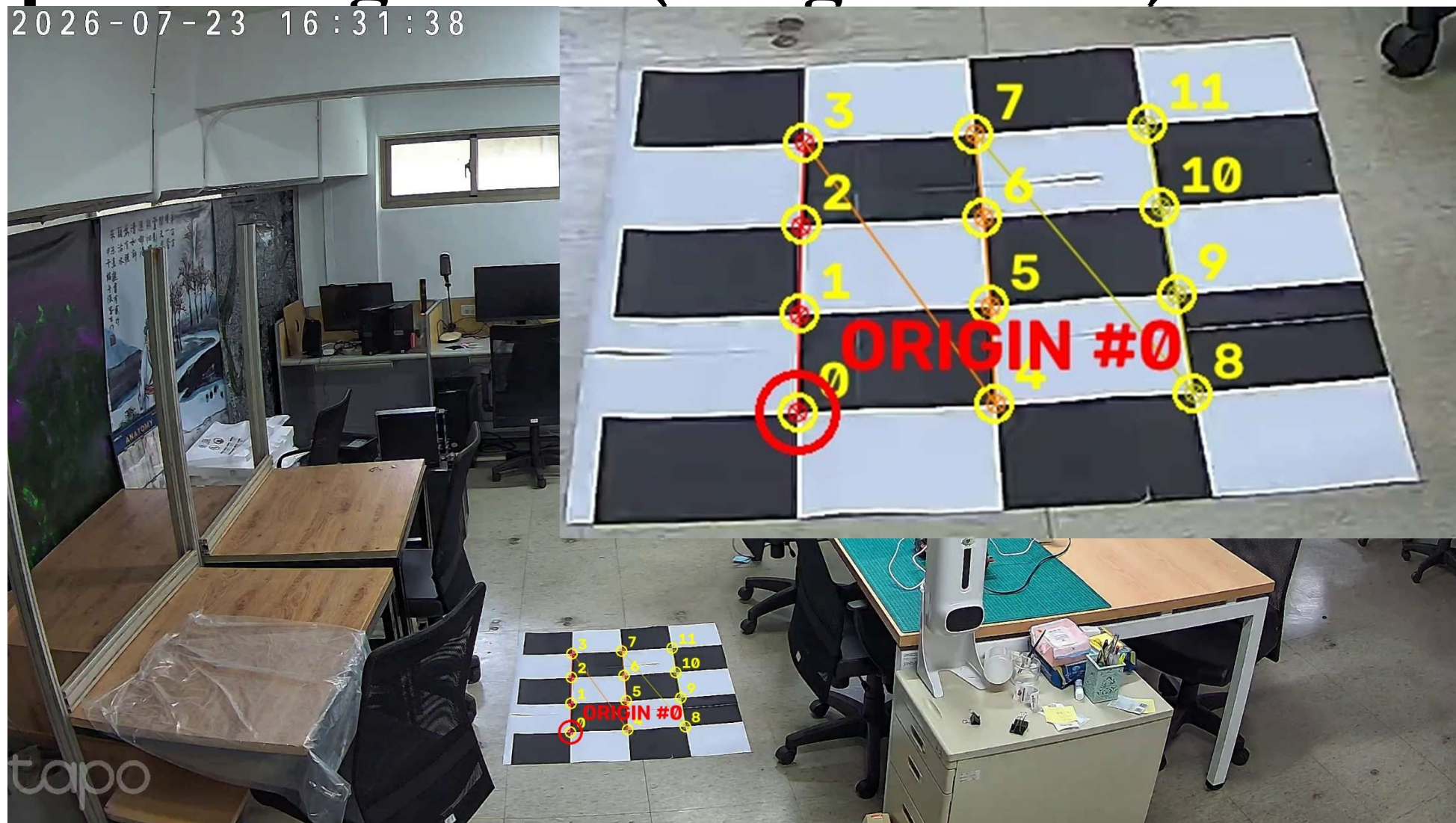
版本	檔案	作法	備註
v1	<code>calibration/homography_v1_manual.json</code>	手動點選磁磚角 (<code>calibrate_boundary.py</code>)	點擊量測誤 差約 8.9 cm
v2	<code>calibration/homography_v2_chessboard.json</code>	地板大棋盤格自動角點 (<code>calibrate_chessboard_floor.py</code>)	目前預設； 點擊量測誤 差約 3.8 cm

Spatial Alignment-(Manual)

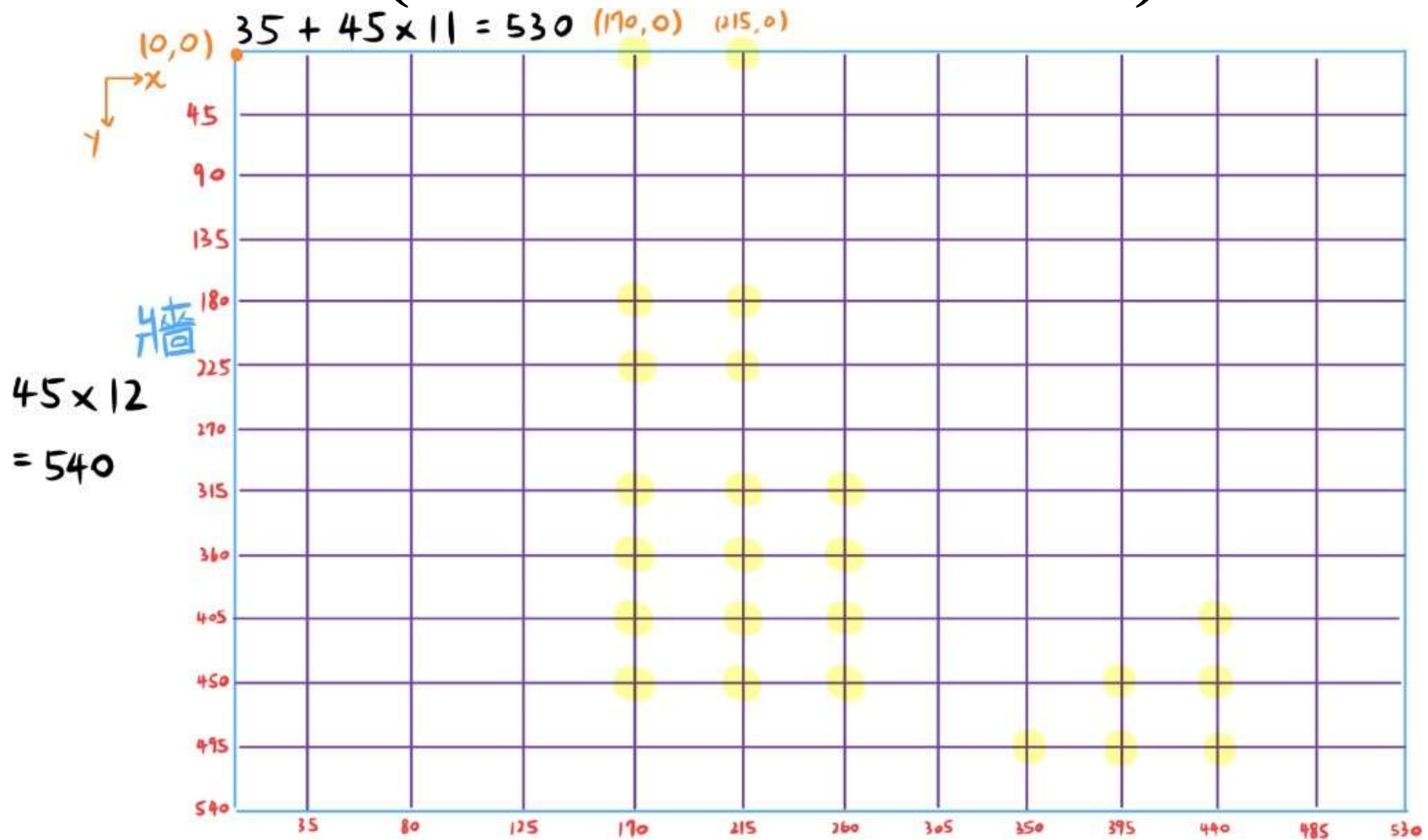


Spatial Alignment-(Target-based)

2026-07-23 16:31:38



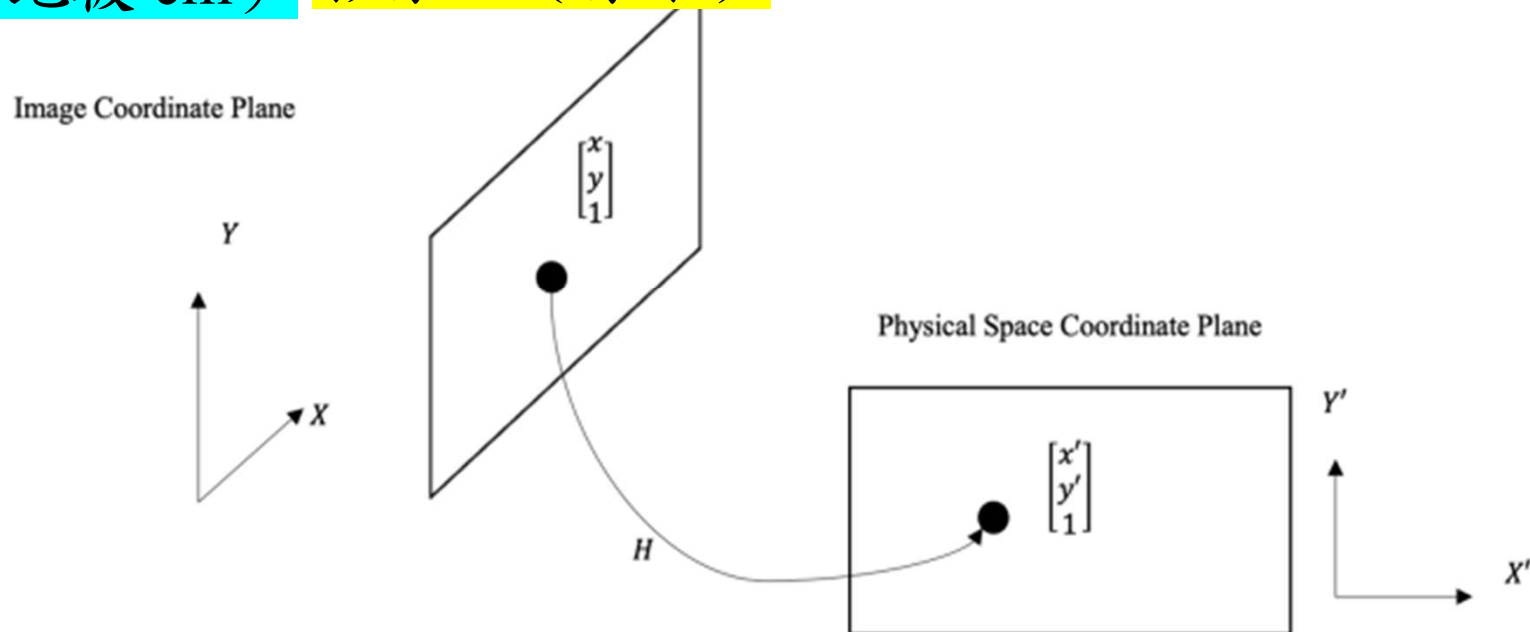
Calibration-(Define Coordinates)



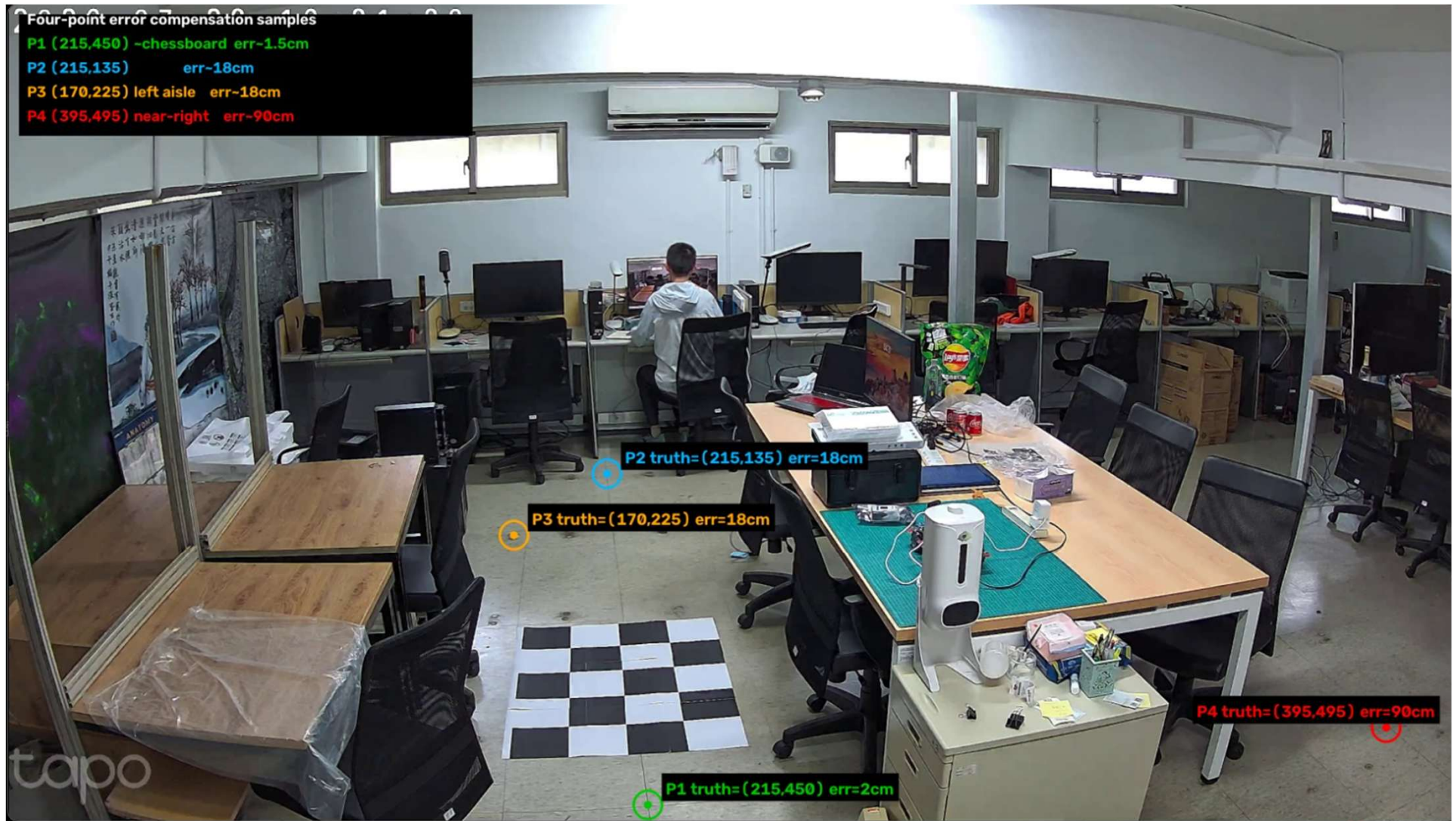
Calibration-(Homography Matrix)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

世界座標 (地板 cm) 影像點 (像素)



Localization Compensation



Stable-ID (Re-ID Re-Identification)

最初時：

地板距離：腳點離某人上次位置，不能瞬移太遠

外貌：人框上半身（軀幹）HSV 顏色直方圖相似度

問題：進出畫面易換號、顏色外貌不穩

目前：

短追蹤：BoT-SORT暫存 raw track

長期 ID：Stable-ID + OSNet-AIN

Gallery：外貌向量 → Cosine Similarity

min-hits = 16：對不上圖庫約連續 16 次偵測才發新號

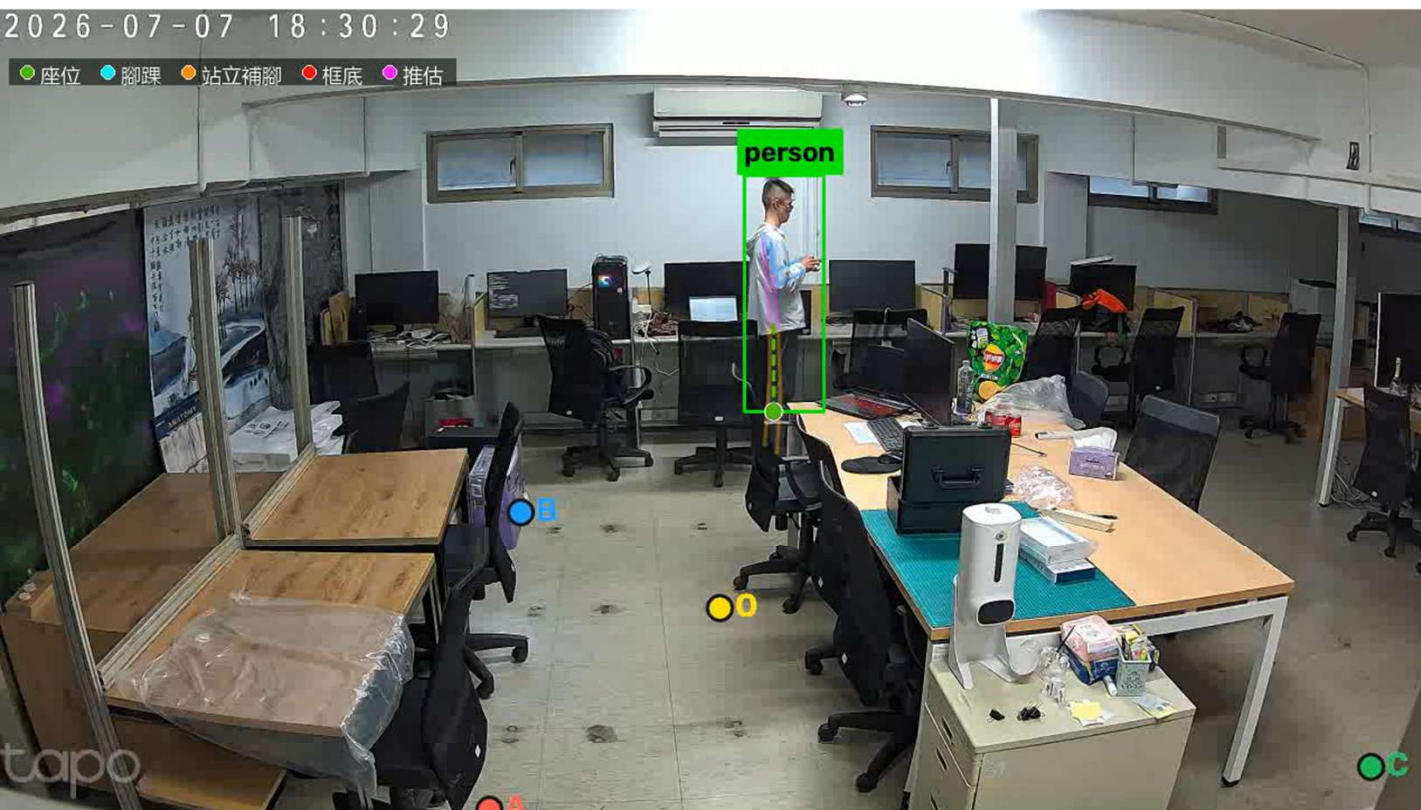
（--stride 5、約 20 fps → 約 4 秒）

YOLO 先偵測人框 → BoT-SORT 做短期追蹤並給臨時 ID
→ Stable-ID 用 OSNet-AIN 看外貌，和 gallery 比對，比
中用舊 ID，比不中且確認後才發新 ID 並存進 gallery。

Result-(test.mp4)

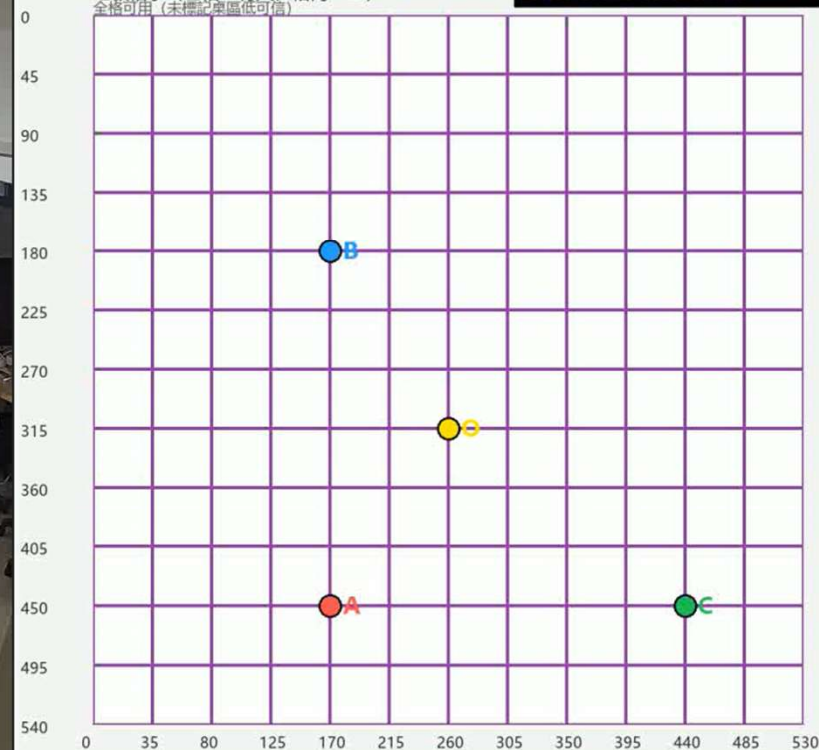
2026-07-07 18:30:29

● 座位 ● 腳踝 ● 站立補腳 ● 框底 ● 推估



地板格子 (色格 = 有人, 格內 = ID)
全格可用 (未標記區域低可信)

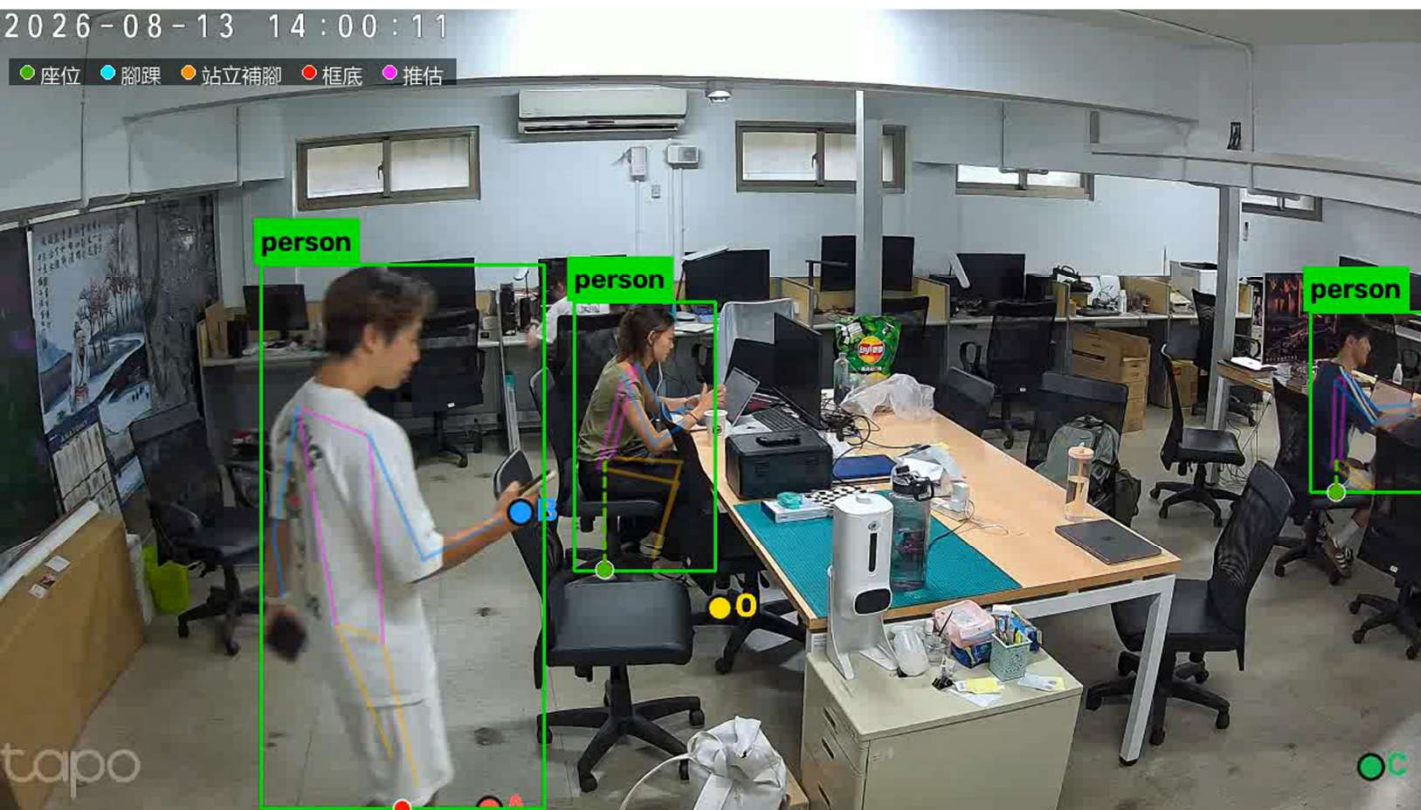
detect 275ms locate 0.42ms



Result-(test4.mp4)

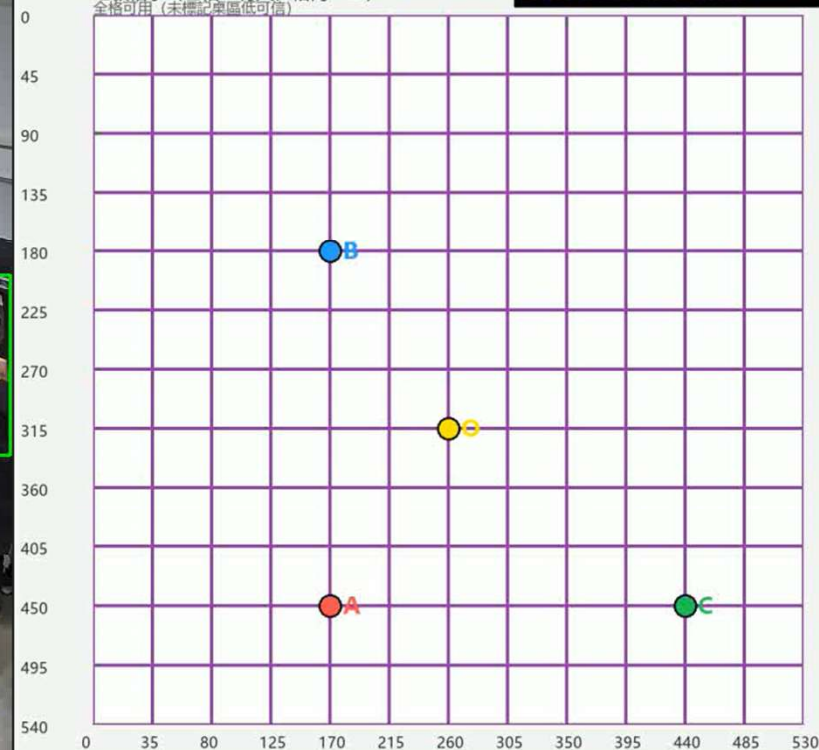
2026-08-13 14:00:11

● 座位 ● 腳踝 ● 站立補腳 ● 框底 ● 推估



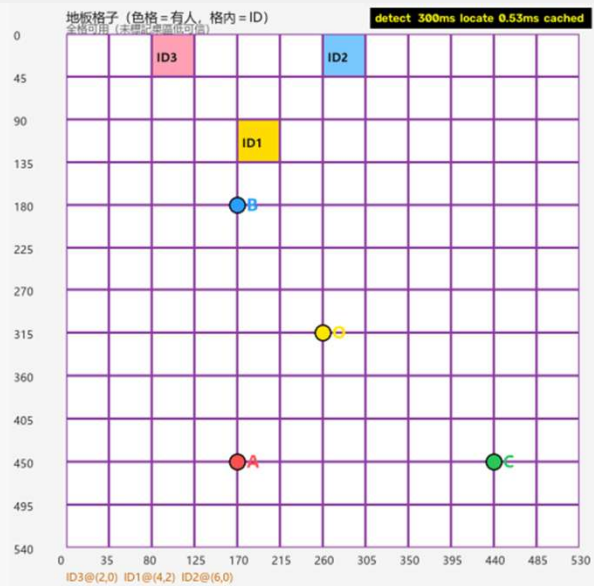
地板格子 (色格 = 有人, 格内 = ID)
全格可用 (未標記區域低可信)

detect 312ms locate 0.85ms



Result-(RTSP)

Grid 格子



Video 監視器



detect_grid launcher GUI

來源

本機影片

RTSP

網址rtsp://oriongo:123456789@192.168.0.200:554/stream1

腳點pose

👤 骨架

👣 追蹤 / ID

📏 四點補償

🔄 A/B/C/O

🔇 少印 log

🖼️ 審查截圖

Re-IDosnet_ain

參數

conf

0.35

stride

5

cell-hold

2

min-hits

16

out-margin (cm)

45.0

格執行

python detect_grid.py --source rtsp://oriongo:123456789@192.168.0.200:554/stream1 --ref pose --conf 0.35 --stride 5 --cell-hold 2 --min-hits 16 --out-margin 45 --quiet --reid-model osnet_ain --error-comp C:\5Gjump\calibration\homography_error_report.json

Run

停止

暫停

本機影片可暫停；按停止結束

[ID-CHANGE] 15:18:13.272 elapsed= 6.88s video= 7.55s frame= 151 → ID1,ID2

Grid 格子

地板格子 (色格 = 有人, 格內 = ID)
全格可用 (未標記地圖皆可值)

detect 279ms locate 1.38ms cached

ID1 @ (4,2) ID2 @ (6,0)

Video 監視器

2026-09-18 15:18:20

● 座位

● 腳蹠

● 站立補脚

● 框底

● 推估

ID1 ID2

B C

A D

感謝您的聆聽